

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

THE LANGEVIN SAMPLER

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Brownian Motion

- Random movement of particles in liquid/gas.
- Robert Brown, botanist: "pollen grains suspended in water perform a continual swarming motion" (1827).

Mathematical model: *a Gaussian process.*

- $B_0 = 0$
- For $t > s$,
 - $B_t - B_s$ is independent of B_s
 - $B_t - B_s \sim N(0, \omega^2(t - s))$
- $t \rightarrow B_t$ is almost surely continuous

Limit of a simple random walk with step size δ and time increment τ going to zero such that $\delta/\sqrt{\tau} \rightarrow \omega$.

Langevin Diffusion

Suppose the target density on $\mathcal{X} = \mathbb{R}^d$ is

$$P(x) \propto e^{-U(x)}.$$

Langevin diffusion (X_t) is defined by stochastic differential equation

$$dX_t = -\nabla U(X_t)dt + \sqrt{2}dB_t$$

where (B_t) is d -dimensional Brownian motion.

Long-term distribution of (X_t) is P .

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How can this process be simulated?

Discretizing the Langevin Diffusion

Euler-Maruyama scheme for sampling diffusion paths:

$$X_{t+1} = X_t - \gamma_{t+1} \nabla U(X_t) + \sqrt{2\gamma_{t+1}} Z_{t+1}$$

where $Z_1, Z_2, \dots$ are i.i.d. $N(0, I_d)$ and γ_t are step sizes.

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- Related to stochastic gradient descent
- If step size is held constant ($\gamma_t = \gamma$):
 - Converges to a unique stationary distribution P_γ
 - But this isn't (necessarily) the same as P
- When step size is decreased: harder to analyze.

Gibbs sampler:

- Formalization of Glauber dynamics in statistical physics.
- Geman, Geman. *Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images*. IEEE Transactions on Pattern Analysis and Machine Intelligence, 1984.
- Gelfand, Smith. *Sampling based approaches to calculating marginal densities*. Journal of the American Statistical Association, 1990.

Langevin sampler:

- Paul Langevin (1872-1946) developed Langevin equation that described evolution of a system under a combination of deterministic and random forces.
- Grenander, Miller. *Representations of knowledge in complex systems*. Journal of the Royal Statistical Society, 1994.
- Besag. *Comments on "Representations of knowledge in complex systems"*. Same journal.